Do Better Generators Make Worse Teachers? Disentangling Diversity, Label Alignment and Fidelity in Synthetic Training Data

Anonymous CVPR submission

Abstract

Newer text-to-image generators produce images that look better yet train worse classifiers, a regression recently attributed to a collapse toward narrow, aesthetic-centric distributions. That diagnosis is correlational, rests on generic ImageNet classes, and comes without a remedy. We ask which property of a synthetic training set actually drives real-test accuracy, and we answer by intervention rather than correlation. Holding the generator, prompt and pool of images fixed, we construct training sets that differ in only one measured property: intrinsic diversity, coverage of the real distribution, label alignment, or per-sample fidelity. We do this in a 2×2 *factorial and in a balanced random-subset design that supports a Shapley decomposition of explained variance. We study four open generators spanning three model generations on two fine-grained domains, CUB-200-2011 and FGVC-Aircraft, where generators know the classes far less well than ImageNet categories. We evaluate with a frozen-backbone linear probe and a from-scratch control. Within a fixed pool, **[coverage / alignment / fidelity]** accounts for **[xx]**% of explained variance in real-test accuracy, against **[xx]**% for fidelity, and FID ranks generators **[worse than / no better than / better than]** random. The decomposition motivates DAS, a training-free selection recipe: oversample, filter by label alignment, then select for coverage. DAS closes **[xx]**% (CUB) and **[xx]**% (Aircraft) of the gap between naive synthetic and matched real data, at no additional generation cost beyond oversampling. We report where it does not help.*

1. Introduction

Text-to-image (T2I) models are an appealing source of training data. They are cheap to query, free of collection cost, and can in principle be asked for any class in any condition [2, 13, 30, 37]. It is natural to expect that as generators improve, the data they produce should improve with them. Adamkiewicz *et al*. [1] showed that the opposite happens. Across thirteen open generators released between 2022 and 2025, classifiers trained only on synthetic ImageNet images and tested on real ones get steadily worse with newer models, even as prompt following and visual quality improve. Their analysis links the regression to degraded texture and high-frequency statistics and to a collapse of the synthetic distribution: high density, low coverage.

This is an important observation, but it leaves the mechanism open in three ways that matter for anyone who wants to use synthetic data. First, the evidence is correlational. Generators differ in architecture, autoencoder, training data, default resolution and guidance behaviour all at once. A cross-generator association between coverage and accuracy cannot separate the effect of coverage from everything else that changed. Second, the evidence concerns 200 generic ImageNet classes that generators have seen countless times. In fine-grained domains such as bird species or aircraft variants, a generator may render a beautiful, diverse image of the wrong species. In that case label alignment, not diversity, could be the binding constraint. Third, no remedy is offered short of obtaining detailed captions of real images, which presupposes the real data that synthetic data was meant to replace.

We address these gaps with a simple change of method: *we intervene on the training set instead of comparing generators.* For a fixed generator and prompt regime we draw a large pool of images and then construct many training sets of identical size from that pool. These sets differ systematically in one measured property while the others are held fixed or balanced. Because generator, prompt, autoencoder and resolution are now constant, differences in real-test accuracy can be attributed to the manipulated property of the set rather than to the model that produced it. We use two designs. One is a 2×2 factorial over diversity and alignment. The other is a balanced random-subset design whose variation supports a Shapley decomposition of explained variance [11, 21].

Taking this route forces several distinctions that correlational studies tend to blur, and we make them explicit. *Intrinsic diversity* (how spread out a set is, measured by the Vendi score [8]) is not the same as *coverage* (how much of the real distribution the set reaches [18, 24]). A generator can be highly diverse along axes that do not matter, such as artistic style. *Label alignment* (does the image show the prompted class?) is not *prompt following* in the GenEval sense, and the predecessor found the latter inversely related

to utility [1]. *Fidelity* measured per sample (precision, density) is not FID, which conflates fidelity and diversity in a single distance [14, 19].

The decomposition suggests a remedy that needs no generator training. DAS (Diversity-matched, Alignment-filtered Sampling) oversamples each class, discards images that a few-shot real classifier or a text-only scorer does not recognise as the prompted class, and then selects for coverage with greedy k-center [10, 31]. We compare it with natural alternatives, including simply training on the whole oversampled pool, so that the benefit of selection is not confused with the benefit of more images.

Contributions.

1. **An interventional decomposition** of the synthetic-to-real gap into diversity, coverage, label alignment and fidelity, with generator and prompt held fixed. It includes an autoencoder-reconstruction control that isolates latent-space artifacts from distributional effects.
2. **Evidence from fine-grained domains** (CUB-200-2011, FGVC-Aircraft) across four generators and three prompt regimes. We show how the relative importance of each factor **[shifts / does not shift]** when generators know the classes poorly, including a per-class analysis that relates the gap to how well the generator knows each class.
3. **DAS**, a training-free selection recipe. We report the fraction of the real-data gap it closes, a compute-matched comparison against using all generated images, and the settings where it fails.
4. **Methodological checks** that we found necessary: a from-scratch control for the linear-probe protocol, two independent alignment scorers, and separation of the embedding used for selection from the one used for measurement, so that DAS cannot win by optimising the metric it is scored on.

We state our hypotheses in advance (Sec. 3.4) and report negative outcomes as findings.

2. Related work

Training on generated images. Early work established that diffusion-generated images can train or pretrain useful classifiers. These include ImageNet clones from Stable Diffusion [30], synthetic data for zero- and few-shot recognition [13], fine-tuned Imagen data that improves ImageNet accuracy [2], and synthetic contrastive pretraining [37, 38]. Fan *et al*. [7] characterised scaling laws for synthetic images and their limits. Geng *et al*. [9] showed that retrieving real images from a generator's own training set beats generating from it. Singh *et al*. [33] benchmarked the robustness of synthetic-trained models. Most of this work uses Stable Diffusion 1.x/2.x [29]. Adamkiewicz *et al*. [1] extended the comparison to thirteen generators and found a regression with newer models, which is the starting point of this paper. Their analysis is correlational across generators on generic classes. Ours intervenes within generators on fine-grained classes and proposes a remedy.

Diversity, prompts and augmentation. Several works observe that synthetic data lacks diversity and attack it at the prompt: by diversifying prompts [32], using captions as prompts [20], or editing real images toward new domains [5, 39]. For fine-grained classes, SaSPA [23] conditions generation on structure to preserve subject identity. DataDream [16] fine-tunes the generator on a few real shots. Real-Fake [42] frames synthesis as distribution matching. These methods change the generator or its conditioning. DAS is complementary: it leaves the generator untouched and changes only which samples are kept, so it can be stacked on top of any of them.

Data selection and label noise. Selecting a coreset by greedy k-center [10, 31] and pruning data by difficulty [34] are established for real data. Filtering by a classifier's agreement with the given label is a standard tool against label noise [25]. In synthetic data a misaligned image is a form of label noise whose rate depends on how well the generator knows the class. We study how filtering and coverage selection interact when the pool itself is collapsed.

Evaluating generative models. FID [14] and KID [3] measure a distance between distributions and therefore mix fidelity with diversity. Inception features are also biased toward ImageNet classes [19]. Precision/recall [18] and density/coverage [24] separate the two. Stein *et al*. [36] recommend DINOv2 features [26] over Inception. The Vendi score [8] measures diversity without a reference set. We use these metrics as candidate explanatory variables and test which ones predict training utility under intervention.

3. Setup, measurements and hypotheses

Notation. A task has K classes, a real training set $\mathcal{D}_r$, and a real test set $\mathcal{D}_{te}$ that is never used for anything except final evaluation. A generator G with prompt regime π produces a pool $\mathcal{P}_{G,\pi}$ of mN images per class (m=3). A training set $S \subset \mathcal{P}$ holds exactly N images per class. The test accuracy of a classifier trained on S is $\mathrm{Acc}(S)$. We fix N to the per-class size of $\mathcal{D}_r$ so that synthetic and real sets have equal size, and we also report curves over N.

We carve a small *alignment split* $\mathcal{A} \subset \mathcal{D}_r$ of k=5 images per class. It is used only to score label alignment and as the training set of the matched real 5-shot baseline. Any method that uses $\mathcal{A}$ is therefore compared against a classifier that is given the same real images.

3.1. Data, generators and prompts

Datasets. CUB-200-2011 [41] (200 bird species, about 30 training images per class) and FGVC-Aircraft [22] (100 variants, standard variant-level split). Both are fine-grained: the classes are distinguished by local cues such as plumage patterns, engine count or winglets, and they are rarer in web captions than ImageNet categories. Dermoscopy (HAM10000 [40]) is reported as an exploratory third domain in the supplement.

Generators. Four open models spanning three generations: SD1.5 [29], SDXL [27], SD3.5-**[Large/Medium]** [6, 35], and FLUX.1-dev [4]. We use **[50]** sampling steps and classifier-free guidance [15] of **[2.0]** wherever the model supports it, following the low-guidance setting known to favour diversity [1, 30]. Two deviations are unavoidable and we report them rather than hide them. FLUX.1-dev is guidance-distilled and exposes a guidance *embedding* rather than true CFG; we set it to **[value]**. SDXL is generated at its native 1024^2; the others at 512^2**[; confirm SD3.5/FLUX]**. All images are resized to 224^2 before training.

Prompt regimes. P1 uses the class name with a domain hint ("a photo of a {class}, a type of bird"). P2 uses LLM-diversified templates that vary pose, background, lighting, season and viewpoint, with the class name always present. The templates are written once per domain without access to real images. P3 uses detailed captions of real training images produced by a vision-language model. We treat P3 as an *oracle* regime, since it leaks information about the real distribution [1]. P1 and P2 are run for all four generators. P3 is run for the two cheapest, SD1.5 and SDXL.

3.2. Measured properties

All metrics are computed per class and averaged, at equal N, because several of them (notably Vendi) depend on sample size. The primary embedding is DINOv2 ViT-B/14 [26], which is also the backbone of the main learner. All metrics are recomputed in CLIP ViT-L/14 [28] space as a robustness check.

Diversity (reference-free). The Vendi score [8]: the exponential of the entropy of the eigenvalues of $\frac{1}{N}\mathbf{K}$, with $\mathbf{K}$ the cosine-similarity kernel of the N embeddings. It is the effective number of distinct samples, and it does not ask whether those samples resemble real data.

Coverage (reference-based). The fraction of real training images whose k-NN ball (k=5) contains at least one synthetic image [24], and improved recall [18]. These ask whether the synthetic set *reaches* the real distribution. The distinction from Vendi is central to our analysis. A set can raise Vendi by spreading into regions real data never occupies, for example painterly styles, without raising coverage at all.

Label alignment. Two independent scorers, both outputting $a(x, y) \in [0, 1]$ for image x prompted with class y. A_{real} is the softmax probability of y under a nearest-class-mean classifier built from the alignment split $\mathcal{A}$ in CLIP image space. A_{text} is CLIP zero-shot probability over the K class names and uses no real images at all. Both are computed in CLIP space, never in the DINOv2 space of the learner, to avoid scoring images with the same features that are later trained on. CLIP zero-shot is known to be weak on aircraft variants. We therefore expect, and report, disagreement between the scorers on FGVC-Aircraft.

Fidelity (per sample). Improved precision [18] and density [24]: how many synthetic samples land on the real manifold. We also report FD in DINOv2 space (FD_{DINO} [36]), Inception FID [14] for comparability, and KID [3]. Because FD-type metrics mix fidelity and diversity, we treat them as *proxies to be tested*, not as one of the factors.

3.3. Learners

Main: frozen DINOv2 linear probe. Multinomial logistic regression on frozen, ℓ_2-normalised DINOv2 ViT-B/14 `[CLS]` features [17]. Weight decay is selected on a held-out *synthetic* validation split. Selecting on real validation data would leak real information into a nominally synthetic-only pipeline, and we report that variant only as an upper bound. A linear probe on a real-pretrained backbone measures how good the images are as *supervision for a head*, not as *pretraining data*. It is fast enough to train thousands of times, which the interventional designs require.

Control: from-scratch ResNet. A ResNet-18/50 [12] trained from random initialisation on a subset of configurations (Sec. 6.3). Absolute accuracies from scratch on about 30 images per class are low. The question we ask of this control is whether it *ranks* training sets the same way the probe does. We report the Spearman correlation between the two learners' accuracies over the shared configurations.

Baselines. Real-only training at $k \in \{5, 10, \text{all}\}$ images per class, and hybrid real+synthetic mixing curves.

Controls for confounds. *Autoencoder ceiling:* we pass $\mathcal{D}_r$ through each generator's VAE (encode then decode, no diffusion) and train on the reconstructions. This bounds how much of the gap is due to latent-space artifacts alone, which affect high frequencies [1]. It costs no diffusion steps. *Contamination:* generators trained on web-scale data, and

DINOv2 trained on curated web images, may have seen images from these benchmarks. We cannot rule this out. We note that it would inflate synthetic and probe accuracy alike, so it biases *toward* synthetic data and against our main conclusions.

3.4. Hypotheses

We fixed these hypotheses before running the interventional experiments.

H1. Within a fixed pool, coverage and label alignment explain more variance in real-test accuracy than per-sample fidelity, and reference-free diversity (Vendi) explains less than coverage.

H2. Across generators, newer models score higher on fidelity and lower on coverage under P1, so FID and precision misrank generators as data sources.

H3. The gap is larger for classes the generator knows poorly. We test this per class by regressing the class-level gap on *class familiarity*, the mean A_{text} of the class under P1. We prefer this within-dataset test to a comparison with numbers reported on ImageNet in other papers, which would differ in protocol.

H4. Alignment filtering followed by coverage selection (DAS) closes a meaningful share of the gap. We commit to reporting the share however small, and to comparing against training on the full oversampled pool.

4. From correlation to intervention

Comparing generators confounds every property of the data with every property of the model. We instead fix (G, π), and with it the autoencoder, resolution, sampler and training corpus, and vary only which N images per class are kept from the pool $\mathcal{P}_{G,\pi}$. The manipulated variable is then the composition of the training set, and its effect on Acc is identified up to properties of the selected images that we did not measure. We return to that caveat in Sec. 7. All selection steps use CLIP embeddings. All diversity, coverage and fidelity *measurements* use DINOv2 embeddings. Selection therefore cannot trivially maximise the quantity it is scored on.

Diversity arm. Fix an alignment threshold τ (the pool's **[40]**th percentile of A_{real}) and restrict to images above it. From the surviving set, choose N per class in one of three ways. *Clustered* takes the N nearest neighbours of a random seed image, giving low diversity. *Random* gives the pool's natural diversity. *k-center* uses greedy farthest-point selection [10], giving high diversity. Mean alignment is held fixed by construction, up to within-threshold variation that we report. k-center is known to favour outliers, which may have lower fidelity. We therefore measure precision in every arm instead of assuming it is constant.

Alignment arm. Fix the selection rule to random and vary the alignment quantile from which images are drawn: bottom, middle or top third of A_{real}. High-alignment images tend to be prototypical and therefore less diverse, so this arm alone does not hold diversity fixed. We correct this by matching. Within each tercile we draw 200 random candidate subsets and keep the one whose Vendi score is closest to the pool median, so that the three alignment levels share a common diversity.

2×2 factorial. Crossing low/high diversity with low/high alignment, constructed as above, gives four training sets per $(G, \pi, \text{dataset})$. With five selection seeds and three training seeds each, the design estimates both main effects and their interaction. An interaction is plausible a priori. Diversity may help only when images are correctly labelled, since spreading out misaligned images adds varied label noise.

Balanced random-subset design and attribution. Factorial cells test two factors at two levels. To attribute variance across all four candidate factors, we additionally draw R=**[500]** subsets per $(G, \pi, \text{dataset})$. Each is formed by mixing random, k-center, clustered and quantile-restricted draws in random proportions. This produces broad and only weakly correlated variation in diversity D, coverage C, alignment A and fidelity F (we report their correlation matrix). We fit

$$\text{Acc}(S) = \beta_D \tilde{D} + \beta_C \tilde{C} + \beta_A \tilde{A} + \beta_F \tilde{F} + \gamma_{G,\pi,d} + \varepsilon, \quad (1)$$

with standardised regressors and a fixed effect γ for each generator–prompt–dataset cell. Variance is therefore explained only *within* pools. We report the Shapley (LMG) share of R^2 for each factor [11, 21]: its average marginal contribution over all orderings of regressors. Unlike standardised coefficients, these shares remain interpretable when regressors are correlated. Confidence intervals come from a cluster bootstrap over subsets and seeds.

We also fit (1) *between* cells, without fixed effects, over the **[10]** generator–prompt configurations per dataset. This is the regression a cross-generator study implicitly runs. It has few points and no control of confounds, so we report it as descriptive and contrast it with the within-pool estimate.

5. DAS: Diversity-matched, Alignment-filtered Sampling

DAS turns the decomposition into a recipe that needs no gradient step on the generator. For each class y:

1. **Oversample.** Generate mN images (m=3) under P2.
2. **Filter.** Discard images with $a(x, y) < \tau$. Here τ is a pool quantile chosen on the *synthetic* validation split, never on real test data. If fewer than N images survive, keep the top-N by alignment and flag the class. We report how often this happens and in which classes.

FIGURE 1 (hook). x: generator (SD1.5 → SDXL → SD3.5 → FLUX), P1.
Left axis: precision and FD_{DINO} (inverted). Right axis: real-test accuracy (linear probe).
One panel per dataset; dashed lines = Real-N and the VAE-reconstruction ceiling per generator.
Error bars: 95% CI over 3 generation seeds × 3 training seeds.

Figure 1. **Looking better, teaching worse?** Per-sample fidelity versus the real-test accuracy of a linear probe trained only on synthetic images, across generator generations under class-name prompts. **[One sentence stating what the figure shows once measured.]**

3. **Select.** From the survivors, choose N images by greedy k-center in CLIP space [10, 31]. The first centre is the survivor with the highest alignment, so that the most reliable image anchors the set.

DAS-real uses A_{real} and therefore the 5-shot alignment split. DAS-text uses A_{text} and no real image at any stage. Selection costs $O(mN^2)$ distance evaluations per class, which is negligible next to generation.

What DAS must beat. *Random-N*: N random images from the pool. *All-mN*: the entire pool. This baseline has three times the images and the same generation cost, so it separates the value of selecting from the value of having more. *Filter-only* and *k-center-only*: each step on its own. *More prompts*: P2 and P3 with random selection. *Real 5-shot*: the same real images DAS-real sees, used directly for training. *Real 5-shot + DAS*: the hybrid.

We summarise each method by the *fraction of gap closed*,

$$\Delta = \frac{\text{Acc}(\text{method}) - \text{Acc}(\text{Random-}N)}{\text{Acc}(\text{Real-}N) - \text{Acc}(\text{Random-}N)}, \qquad (2)$$

where Real-N is the real training set at matched size. Δ is undefined when the denominator is near zero, and in that case we report raw accuracies only. $\Delta>1$ would mean the method surpasses real data of equal size.

6. Results

6.1. Cross-generator picture

Figure 1 and Tab. 1 report the uncontrolled comparison that motivates the paper. It replicates, on fine-grained data, the experiment of [1].

[*Result to write:* Does accuracy fall with newer generators under P1 on CUB and Aircraft? Report the monotonicity (Kendall τ over release order) with CI. State whether precision and FD_{DINO} rank generators in the same or the opposite order as accuracy (H2). If the regression does *not* replicate on fine-grained data, say so plainly; that is a finding about domain dependence, not a failure of the paper.]

[*Result to write:* How much does P2 change the ranking relative to P1, and how close does oracle P3 get? Compare each generator to its VAE ceiling. The gap between Real-N and VAE-reconstructed real data is the part that *no* selection method can remove.]

6.2. Interventional decomposition

[*Result to write:* Report the main effects of diversity and alignment and their interaction from the factorial (Fig. 2a), per dataset, with CIs. Is the diversity effect present only under high alignment, as the label-noise argument in Sec. 4 predicts?]

[*Result to write:* Report the within-pool Shapley shares (Fig. 2b) of D, C, A, F with CIs and the model's R^2. Test H1 directly: is $C+A$ larger than F, and is C larger than D? Then contrast with the between-generator decomposition. If the two disagree, this is the paper's central methodological point: cross-generator correlations attribute to diversity what is, under intervention, **[alignment / fidelity / something else]**. If they agree, the interventional design *confirms* the predecessor's diagnosis on new domains, which is still worth stating.]

[*Result to write:* H3: report the per-class regression of the gap on class familiarity, with slope, CI and R^2 per dataset, plus a scatter in the supplement. Name a few of the least familiar classes; if they share a cause (*e.g.* rare sub-variants, visually near-identical sibling classes), say so.]

Robustness of the decomposition. [*Result to write:* (i) Shares recomputed with all metrics in CLIP instead of DINOv2 space. (ii) A_{text} instead of A_{real}, and the rank agreement between scorers, expected to be weaker on Aircraft. (iii) Pairwise correlations of D, C, A, F across subsets. High collinearity limits what any decomposition can claim, and we say so if it occurs.]

6.3. From-scratch control

[*Result to write:* On the subset of configurations trained from scratch (ResNet-**[18]**, **[epochs]**, **[resolution]**), report the Spearman correlation with probe accuracy over shared training sets, and whether the factorial's main effects keep their sign. A low correlation would mean conclusions drawn with the probe do not carry over to synthetic data used as pretraining data. We would then restrict our claims to the supervision setting.]

6.4. DAS

[*Result to write:* State Δ for DAS-real and DAS-text on both datasets with CIs, and whether DAS beats All-mN. This is the comparison that separates a useful selection rule from merely generating more. Report whether DAS-real beats

		CUB-200-2011					FGVC-Aircraft				
Training data	Prompt	Acc↑	Vendi↑	Cov.↑	Prec.↑	A_{real} ↑	Acc↑	Vendi↑	Cov.↑	Prec.↑	A_{real} ↑
Real, 5/class	–	[]	–	–	–	–	[]	–	–	–	–
Real, 10/class	–	[]	–	–	–	–	[]	–	–	–	–
Real, all (N)	–	[]	[]	–	–	–	[]	[]	–	–	–
SD1.5	P1	[]	[]	[]	[]	[]	[]	[]	[]	[]	[]
	P2	[]	[]	[]	[]	[]	[]	[]	[]	[]	[]
	P3 †	[]	[]	[]	[]	[]	[]	[]	[]	[]	[]
SDXL	P1	[]	[]	[]	[]	[]	[]	[]	[]	[]	[]
	P2	[]	[]	[]	[]	[]	[]	[]	[]	[]	[]
	P3 †	[]	[]	[]	[]	[]	[]	[]	[]	[]	[]
SD3.5	P1	[]	[]	[]	[]	[]	[]	[]	[]	[]	[]
	P2	[]	[]	[]	[]	[]	[]	[]	[]	[]	[]
FLUX.1-dev	P1	[]	[]	[]	[]	[]	[]	[]	[]	[]	[]
	P2	[]	[]	[]	[]	[]	[]	[]	[]	[]	[]

Table 1. **Main results.** Real-test top-1 accuracy (%) of the frozen DINOv2 linear probe trained on N random images per class, with set-level measurements in DINOv2 space (alignment in CLIP space, Sec. 3.2). Mean over seeds; 95% CIs and FD_{DINO}/FID/KID/recall/density in the supplement. †P3 is an oracle that captions real training images.

FIGURE 2a. 2×2 factorial. Grouped bars: real-test acc. for {low,high} diversity × {low,high} alignment, one group per generator (P2), one row per dataset. Annotate main effects and interaction with 95% CI.

FIGURE 2b. Shapley share of R^2 for D, C, A, F. Two stacked bars per dataset: *within-pool* (Eq. 1, with fixed effects) vs. *between-generator* (no fixed effects).

Figure 2. **Which property of a synthetic training set matters?** (a) Manipulating diversity and alignment from a fixed pool. (b) Variance decomposition over **[R]** subsets per pool, contrasted with the cross-generator regression. **[One-sentence takeaway once measured.]**

	Real	CUB		Aircraft	
Method	imgs	Acc	Δ	Acc	Δ
Random-N	0	[]	0	[]	0
All-mN (3× data)	0	[]	[]	[]	[]
k-center only	0	[]	[]	[]	[]
Filter only (A_{text})	0	[]	[]	[]	[]
DAS-text	0	[]	[]	[]	[]
Filter only (A_{real})	5	[]	[]	[]	[]
DAS-real	5	[]	[]	[]	[]
Real 5-shot	5	[]	[]	[]	[]
Real 5-shot + DAS-real	5	[]	[]	[]	[]
P3 † + Random-N	all	[]	[]	[]	[]
Real-N	all	[]	1	[]	1

Table 2. **DAS versus baselines** (P2, generator **[best/averaged]**; per-generator results in the supplement). Δ: fraction of gap closed (Eq. (2)). "Real imgs": real images per class used anywhere in the pipeline.

Real 5-shot. If it does not, the honest conclusion is that five real images are worth more than a curated synthetic set, and the hybrid row becomes the practical recommendation. Report the per-generator spread: does DAS help newer generators more, as expected if their deficit is coverage rather than alignment?]

[*Result to write:* Ablations (supplement, one sentence each here): $m \in \{2, 3, 5\}$; threshold quantile τ; selection embedding (CLIP vs. DINOv2, where DINOv2 selection is expected to inflate the probe's score); first-centre rule.]

Mixing with real data. [*Result to write:* From Fig. 3: the real-data budget at which synthetic data stops helping or starts hurting, and whether DAS moves that crossover point.]

7. Discussion and limitations

What the interventions do and do not identify. Selecting from a fixed pool removes confounds that cross-generator comparisons cannot: architecture, autoencoder, training data and sampler. It does not make the four measured factors exhaustive. Any unmeasured property that co-varies with our selection rules is absorbed into the factor it travels with.

FIGURE 3. Mixing curves. x: real images per class {0,5,10,all}; y: real-test acc.; lines: real only, real + Random-N, real + DAS-real. One panel per dataset; generator **[which]**.

Figure 3. **Does synthetic data still help once real data is available? [Takeaway: the real-data budget beyond which adding synthetic images stops helping, if any.]**

FIGURE 4. Qualitative failure modes, 2 rows × 6 synthetic images. Row 1: within-class collapse (FLUX, P1; near-identical pose/background). Row 2: misaligned samples removed by the filter (wrong species/variant), each annotated with A_{real} and the real-classifier's top label. Synthetic images only; no real-image reproduction needed.

Figure 4. **What the filter and the selector remove. [Caption once examples are chosen; pick examples by a stated rule (*e.g.* lowest-A_{real} in 3 random classes), not by hand, to avoid cherry-picking.]**

Examples are texture statistics, which the predecessor found degraded [1], or background clutter. The decomposition is therefore a statement about which *measured* property, when manipulated, moves accuracy. It is not a complete causal account of why a generator's pool looks the way it does. The autoencoder ceiling bounds one important unmeasured cause. Others remain.

Scope of the learner. The probe measures synthetic images as supervision for a head on a strong real-pretrained backbone. That is how practitioners most often use synthetic data in low-data regimes, but it is not synthetic pretraining. The from-scratch control tests whether the ranking transfers (Sec. 6.3). We confine our claims to whatever it supports.

Scope of the data. Two fine-grained domains, four open generators and classification only. Closed generators, detection and segmentation, and generators fine-tuned on in-domain data [16] are out of scope. DAS should compose with the latter, but we have not shown that. DAS-real needs five real images per class. DAS-text needs none, but its scorer inherits CLIP's weaknesses on fine-grained classes. On domains CLIP knows poorly, a text-only filter may discard correct images and keep wrong ones.

Metric dependence. Every set-level metric depends on an embedding. We separate the selection embedding (CLIP) from the measurement embedding (DINOv2) and recompute everything in the other space. Still, no embedding is neutral, and conclusions that flip between spaces should not be trusted. We mark any that do.

Implications. [*Result to write:* One paragraph, written last, stating what a practitioner should do differently. For example: which metric to monitor instead of FID; whether to spend compute on more prompts, more samples, or a five-image real alignment set; and what generator developers should measure if they want their models to be useful data sources.]

Data, licences and societal impact. We use only existing, published datasets (CUB-200-2011, FGVC-Aircraft and, in the supplement, HAM10000) under their research-use terms. We collect no new human-subject data. HAM10000 contains de-identified dermoscopic images released by its creators. We use it only to probe generator limits and make no clinical claim. Generators are used under their respective licences for non-commercial research. The main foreseeable risk is over-confidence. A selection recipe that narrows the synthetic-to-real gap on benchmarks could encourage substituting synthetic for real data in high-stakes domains such as medical imaging, where generator failures are hardest to see. Our results, including the domains where DAS does not help, argue for validating on real data in exactly those settings. Code and selection indices will be released.

LLM Usage Statement

Manuscript. An LLM assistant (Claude, Anthropic) was used to draft the structure and initial text of all sections from the authors' research plan, and to compile the bibliography. The authors verified every citation against the original publication, rewrote the text, and are responsible for all claims. **[Describe any further use for editing or grammar, and name the tool.]** **Research.** The P2 prompt templates (Sec. 3.1) were generated with **[LLM name/version]** from a fixed instruction, reproduced in the supplement. Detailed captions for P3 were produced by **[VLM name/version]**. **[State whether any LLM was used for code, analysis or figures, and to what extent; if none, say so.]** No LLM was used to generate, alter or select experimental results.

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